# Simple and compound gear trains

National 5 · Mechanisms & structures · Answers

## Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

**1a) Calculate the velocity ratio.**
- 45 ÷ 15 = 3

**1b) Calculate the speed of B.**
- 900 ÷ 3 = 300 revs $min^{-1}$

**2a) Calculate the velocity ratio.**
- 60 ÷ 20 = 3

**2b) Calculate the output speed.**
- 720 ÷ 3 = 240 revs $min^{-1}$

**3a) Calculate the velocity ratio.**
- (75 ÷ 25) × (80 ÷ 20) = 3 × 4 = 12

**3b) Calculate the output speed of D.**
- 1000 ÷ 12 = 83 revs $min^{-1}$

**4a) Calculate the velocity ratio.**
- 40 ÷ 10 = 4

**4b) Calculate the output speed.**
- 600 ÷ 4 = 150 revs $min^{-1}$

**4c) State the direction of the output compared with the driver.**
- Opposite direction

**5a) Calculate the velocity ratio.**
- (64 ÷ 16) × (72 ÷ 24) = 4 × 3 = 12

**5b) Calculate the output shaft speed.**
- 1500 ÷ 12 = 125 revs $min^{-1}$

**6a) Calculate the velocity ratio.**
- B is the driven gear: VR = 10 ÷ 30 = 0.33 (1 : 3)

**6b) Calculate the speed of B.**
- 300 × 3 = 900 revs $min^{-1}$

**7a) Calculate the velocity ratio.**
- (72 ÷ 18) × (60 ÷ 24) = 4 × 2.5 = 10

**7b) Calculate the output speed.**
- 1200 ÷ 10 = 120 revs $min^{-1}$

**8a) Calculate the velocity ratio.**
- 36 ÷ 12 = 3

**8b) Calculate the speed of the driven gear.**
- 1800 ÷ 3 = 600 revs $min^{-1}$

**8c) State one reason for using plastic gears in the model.**
- Cheaper, lighter or self-lubricating (less friction and heat)

**9a) Calculate the overall velocity ratio.**

- (70 ÷ 14) × (80 ÷ 20) = 5 × 4 = 20

**9b) Calculate the motor speed needed.**

- 20 × 20 = 400 revs $min^{-1}$